Fiscal Year:	FY 2009	**Task Last Updated:**	FY 02/27/2009
PI Name:	Mitchell, Cary Ph.D.		
Project Title:	(NSCORT) for Advanced Life Support at Purdue/Howard/AAMU - Air and Water Resource Recovery		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	West Lafayette	**State:**	IN
Zip Code:	47907-2010	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSCORT
Start Date:	12/01/2002	**End Date:**	11/30/2009
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 11/30/2009 per PI (1/09) NOTE: Received NCE to 11/30/2008 per PI (11/07)		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Banks, Katherine (Purdue University) Jones, Kimberly (Howard University) Blatchley, Ernest (Purdue University) Heber, Al (Purdue University)		
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Resource Recovery (R2) Executive Summary The NSCORT Resource-Recovery (R2) research group consisted of nine projects with 11 principal investigators. In the four years of Center operation, the research investigators have successfully completed a number of project objectives. Highlights of our research accomplishments are summarized as follows. The “Proof of Concept” was established for the simultaneous treatment of graywater and gas loaded with ammonia and hydrogen sulfide in the BREATHe I prototype. Key design parameters such as size, recirculation rates, media type and flow rates for BREATHe I were optimized. The biodegradability of surfactants was assessed, surfactant degradation byproducts were screened and metabolite biodegradation potential analyzed. In BREATHe II, the feasibility of biofiltration technology was established for treating multi-component contaminated air representative of major cabin air trace contaminants during long duration space missions. Desirable biofiltration operating strategies in terms of packing media type and bioreactor configuration were recommended. A mathematical model was developed for the BREATHe II process. A trace contaminant lab established in year 4 was set up for on-line evaluation of biofilter outlet gases. Independent and combined effects of UV radiation and iodine on water disinfection using Bacillus subtilis spores as the challenging microorganism have been investigated. Research endeavor in membrane processes in ALS contributed to the understanding of how the MF pretreatment membrane and NF polishing membrane can be modified and synthesized to improve wastewater recovery efficiency and mitigate membrane fouling. Our NSCORT Resource Recovery group has maintained a high-level of collaborative interaction throughout the 4-year research period, which is essential since the projects are significantly interrelated. Also, significant research collaborations between NASA field center scientists and NSCORT R2 group PIs were established. Student involvement with our research projects has been extensive and has resulted in the training and graduation of numerous graduate and undergraduate students. Research findings from this group have resulted in a plethora of publications both in conference proceedings and peer-reviewed journals. In conclusion, our group has successfully completed the resource recovery research projects and made significant progress toward NSCORT research objectives. Al Heber, R2 Focus Area Lead
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Development of new water and air-treatment technologies for environmental protection Will impact air-treatment technology and air quality control strategies used in residential and commercial buildings and in commercial aircraft. Further development of fundamental photochemical reactor theory as it relates to systems used for disinfection of water and air.
Task Progress:	Development of a System for Disinfection of Potable Water Based on the Combined Application of UV Radiation and Iodine: UV DISINFECTION SYSTEM FOR CABIN AIR (Dr. Ernest R. Blatchley III, Purdue University) In human-inhabited environments that incorporate recirculation and reuse of water and air, control of pathogenic microorganisms is critical. Ultraviolet (UV) radiation has been demonstrated to be effective for disinfection of water in these settings, in that UV-based systems provide broad-spectrum antimicrobial behavior with minimal disinfection by-product formation; these systems also have low requirements for power and space. These same attributes suggest that UV-based systems may be effective for control of airborne microbial pathogens. Therefore, the objective of this research is to develop a system for UV disinfection of recirculating air. This ongoing project involves three basic components. First, existing literature information suggests that the UV dose-response behavior of airborne microorganisms is influenced by relative humidity; however, the experiments that have reported this behavior to date have been based on crude systems that do not allow for precise determination of several critical factors in the experiment, including UV dose delivery. We have developed an experimental method that allows for precise measurement of UV dose delivery, as well as the response of microorganisms to UV exposure. Moreover, this system allows for precise control of relative humidity and temperature, so that unambiguous results can be obtained. This system will be used to characterize and quantify UV dose-response behavior of several relevant microorganisms as a function of relative humidity. This dose-response data will then be used as input to a numerical model we will develop based on the combined application of computational fluid dynamics (CFD) and UV intensity (I) field models. We have used this CFD-I modeling approach successfully to design a highly efficient UV-based reactor for disinfection of water; the application to air disinfection is a natural extension of this modeling approach. The CFD-I modeling work will allow us to identify an efficient reactor design through the process known as numerical prototyping. Based on these results, a physical prototype of the reactor will be built and constructed to validate the reactor’s performance, as well as the predictions of the CFD-I model. This work will be done in partnership with Trojan Technologies, Inc, of London, Ontario, Canada. Trojan Technologies is one of the leading manufacturers of UV-based treatment systems, and they are interested in participating in the development of this system for air treatment. In addition, engineers at Airbus in Europe have expressed an interest in the development of these systems for disinfection of cabin air on commercial aircraft. We anticipate that the results of this work will yield three manuscripts that will be submitted for publication in refereed journals. Some aspects of this work were presented at COSPAR in Montreal in the Summer 2008; we anticipate that other aspects of the work will also be presented at professional meetings. The Ph.D. student who is working on this project, Mr. Soojung Lim, is expected to complete his degree by November 2009.
Bibliography Type:	Description: (Last Updated: 04/16/2020)

Articles in Peer-reviewed Journals	Naunovic Z, Pennell KG, Blatchley ER 3rd. "Development and performance of a fluence rate distribution model for a cylindrical excimer lamp." Environmental Science Technology 2008 Feb;42(5):1605-14. http://dx.doi.org/10.1021/es070968w , Feb-2008
Articles in Peer-reviewed Journals	Sharvelle S, McLamore E, Banks MK. "Hydrodynamic characteristics in botrickling filters as affected by packing material and hydraulic loading rate." Journal of Environmental Engineering 2008 May;134(5):346-52. , May-2008
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Articles in Peer-reviewed Journals	Sharvelle S, Arabi M, McLamore E, Banks MK. "Model development for biotrickling filter treatment of graywater simulant and waste gas. I." Journal of Environmental Engineering 2008 Oct;134(10): 813-25. http://dx.doi.org/10.1061/(ASCE)0733-9372(2008)134:10(813) , Oct-2008
Articles in Peer-reviewed Journals	Sharvelle S, Arabi M, Banks MK, Mannering F. "Model sensitivity analysis for biotrickling filter treatment of graywater simulant and waste gas. II." Journal of Environmental Engineering 2008 Oct;134(10): 826-34. http://dx.doi.org/10.1061/(ASCE)0733-9372(2008)134:10(826) , Oct-2008
Articles in Peer-reviewed Journals	Pennell KG, Naunovic ZZ, Blatchley ER 3rd. "Sequential inactivation of Bacillus subtilis spores with UV radiation and iodine." Journal of Environmental Engineering, 2008 Jul;134(7):513-20. http://dx.doi.org/10.1061/(ASCE)0733-9372(2008)134:7(513) , Jul-2008